sci

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

WHAT'S NEW

Silk coated seeds can be used to grow crops in salty soil

Researchers from MIT have found a way to use the ordinary silk produced from silkworms to make cocoons for seeds. Bacteria and nutrients can be embedded within the silk cocoons, providing nutrients to the seeds. The approach could allow the growing of crops in salty soil which is not suitable for agriculture. Without the silk protection, the salt would prevent the seeds from growing. The process for coating the seeds with silk is inexpensive, and does not require any specialised equipment. The silk, along with sugar and bacteria are all suspended in water, and the seeds only need to be soaked in the water

for a few seconds to acquire an even coating. Another approach is to use a specialised spray to provide the coating to the seeds. Both the processes are cheap, easy, fast and scalable to handle large amounts of seeds. The approach has the added environmental benefit of requiring less or no use of artificial fertilizers, the production of which is expensive and damaging to the environment as well.
https://dgit.in/slksd

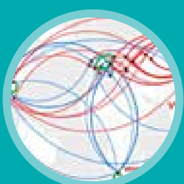
How bees escape falling into water

A Caltech research engineer just happened to see a bee struggling to escape from a water fountain, leading to some incredible science. The movement of the wings of the bee created waves in the fountain, the amplitude of which could be seen in the shadows. There was also an interference pattern generated as the waves generated by a particular wing beat crashed into the waves generated by another.

The researcher "rescued" the honeybee and got it back to the lab to study the behaviour further. Working together with another aeronautics and bioengineering researcher, the team studied the movements of bees as they struggled to escape from a shallow pan of perfectly still water. The bee navigates to the edge with minimal effort, allowing it to surf in a particular direction on the waves of its own creation. At the edge of the water, it can pull itself out and fly away. A paper describing the findings of the team has been published in the Proceedings of the National Academy of Sciences.
https://dgit.in/waterbee

THIS MONTH IN SCIENCE:

This month in Earth, we cover Cryoseisms which is the vibrations in the ground caused by ice, then in Health, we take a deeper look at sweat sensors. In Space Age, we learn about spacecraft design inspirations taken from nature and in Ignorance, we tackle Baryon assymetry



Mapping the spread of hate

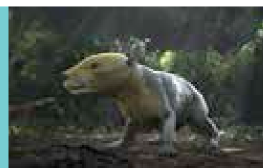
A first of its kind study tries to understand the inner workings of the internet h8 machine, by forming self organised and scalable clusters across geographic locations, that then interconnect to form resilient networks across multiple social media platforms.
https://dgit.in/ih8m

Beyond Moore's Law

The ICs used in computers are approaching the limits of transistor densities, at least in the two dimensional arrays that they are typically arranged in. A new approach of using three dimensional arrays can significantly improve computing power.
https://dgit.in/3dic

Stellar slingshot

About five million years ago, a star had a close encounter with the supermassive black hole at the center of the Milky Way, and was hurled outwards at an incredible speed as a result. The star is currently moving through the galaxy at ten times the speeds of other stars.
https://dgit.in/starshot



A fossil trove of recovery

Incredibly well-preserved fossil finds in the Denver basin documents the revival of life in the hundred million years following the asteroid impact that is linked to the extinction of the dinosaurs. The fossils have remains of both mammals and dinosaurs.
https://dgit.in/fsltrv